

Bin Location **Allocation with Weight Restriction**

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the **Allocation with Weight Restriction in Bin Locations** course topic.

You must have a good knowledge of inventory, UoM and bin location management in order to fully understand the content of this course.

Objectives



At the end of this topic, you will be able to:

- Assign and track weight for items and bin locations
- Enable allocation to bin locations according to weight restriction

After completing this training, you will be able to:

- Assign and track weight for items and bin locations
- Enabling allocation to bin locations according to weight restriction

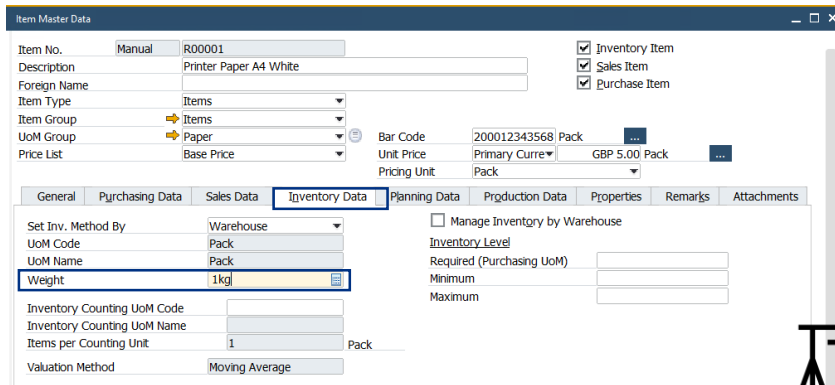
Business Scenario



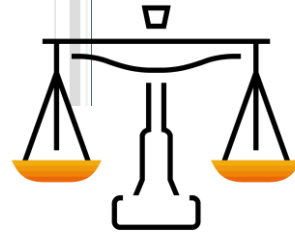
- OEC Computers resell computer and electronic equipment to retailers in the UK.
- OEC Computers have already implemented the bin locations and UoM solution in the company.
- George, the warehouse manager, would like to start manage weight for the items in the warehouse. A weight indication per item will allow him to monitor the weight of items in storage and limit the weight for carrying and loading goods up to the standard allowed.
- George, has asked you, his SAP Business One consultant, to help him manage weight for items and bin locations in the warehouse.

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Weight Indicator in the Item Master Data



The screenshot shows the SAP Item Master Data window for item R00001, 'Printer Paper A4 White'. The 'Inventory Data' tab is active. The 'Weight' field is highlighted with a blue box and contains the value '1kg'. The 'UoM Code' is 'Pack' and the 'UoM Name' is 'Paper'. The 'Valuation Method' is 'Moving Average'. The 'Bar Code' is '200012343568' and the 'Unit Price' is 'GBP 5.00 Pack'. The 'Pricing Unit' is 'Pack'. The 'Inventory Item', 'Sales Item', and 'Purchase Item' checkboxes are all checked.



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Let us start by looking at the *Weight* field in the *Item Master Data*. This is the weight of the inventory UoM of the item.

In the image we can see the **Printer Paper** item. We can see that this item is related to the **Paper** group and that 1 Pack of **Printer Paper** equals **1** kilogram.

Note that each time we update the *Weight field* of the *Inventory Data*, the system prompts a message asking if we want to copy the inventory UoM weight to the sales and purchasing RUoM. The weight is then copied proportionally, according to the conversion rules defined between the different UoM codes in the UoM group. This functionality allows us to update all three weight field at once.

Inventory UoM Weight Indicator – Business Example

Monitor Inventory UoM weight to avoid overweight in carrying and loading goods

- A forklift weight limitation = 500 kg
 - 1 pack of paper = 1 kg
-
- A forklift can carry 500 packs of printer paper



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Setting the weight for the inventory UoM of the item allows us to monitor and track item weight in the warehouse.

It can also be used to limit the weight lifted and carried for the warehouse employees or warehouse machinery.

In the example shown here, we can see that the forklift has a weight limitation of **500 kg**.

- Since we manage weight for the inventory UoM of the item, we can easily calculate how many items we can load on the forklift without having to weigh the items each time.
- 1 pack of printer paper weighs **1 kg** and this means that a forklift can carry **500 packs** of **printer paper**.

Allocation by Maximum Weight - Concept

Allocation by weight principles:

- ✓ We can define weight for the inventory UoM of the items
- ✓ We can define weight capacity for each bin location
- ✓ We can then choose to restrict item allocation according to the bin location weight capacity.



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The bin location solution allows different automatic allocation rules.

This solution includes an allocation restriction by weight.

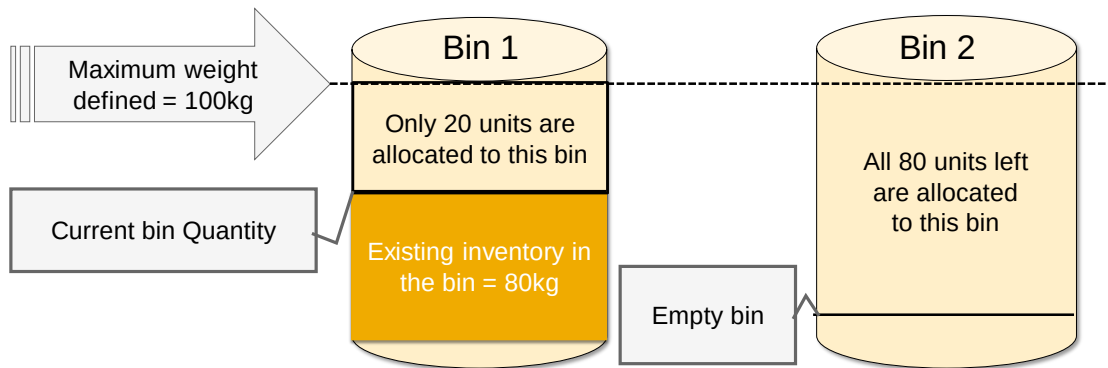
We can define for each bin location its weight capacity.

We also already saw that we can define the weight of the inventory UoM of the item.

These definitions enable performing a bin location allocation up to the maximum allowed for each bin.

Allocation by Maximum Weight - Example

- Maximum weight restriction for each bin = 100 Kg
- 50 hard disk units to allocate. Each hard disk weighs 1 kg => A total of 50 Kg to allocate



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Look at the example demonstrated here.

We want to allocate hard disk items. Each hard disk weighs **1 kg**.

We have 2 bin locations: Bin 1 and Bin 2. Each has a weight restriction of 100 kg. Bin 1 has already an existing inventory of **80 kg**. Bin 2 is empty.

There is also an allocation rule. According to this rule, allocation is done to Bin 1 first and then to Bin 2.

We can see that when allowing weight restriction to a bin, the system will allocate only **20 kg** to Bin 1 and the rest to Bin 2.

Warehouse Setup - Maximum Weight Allocation Definition

Set weight allocation rules in the Warehouse Setup window:

The screenshot shows the 'Warehouses - Setup' window with the 'Bin Locations' tab selected. The 'General' section is visible, showing fields for 'Bin Loc. Code Separator' (set to '-'), 'No. of Bin Locations' (94), and 'Default Bin Location' (05-A001-S1). The 'Bin Locations' section includes checkboxes for 'Enable Receiving Bin Locations' (checked), 'Restrict Receipts to Empty Bin Locations' (checked), and 'Receive up to' (set to 'Maximum Weight'). A callout box labeled 'Automatic allocation rules' points to the 'Receive up to' dropdown. Another callout box labeled 'Receiving Bin Locations' points to the 'Maximum Weight' option in the dropdown. The dropdown menu also shows 'Maximum Qty' and 'Max. Qty and Weight' options.

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In the *Warehouse – Setup* window we indicate if we want to use the rule of allocation up to the maximum weight.

We can define this rule for both Receiving and regular non-Receiving bin locations.

For regular bin locations we can add a weight validation in addition to any allocation rule on receipt. Note that this definition is relevant for both automatic and manual allocation.

For Receiving bin location, we can add a validation for the maximum quantity, weight or quantity and weight. Both maximum weight and maximum quantity are defined in the *Bin Location Master Data*. Let us it in the next slide.

Weight Fields in the Bin Location Master Data

The screenshot shows the 'Bin Location Master Data' form in SAP. The form includes fields for Warehouse (05), Aisle (SYSTEM-BIN-LOCATIO), Area (S1), Row (L1), and Shelf. The Bin Location Code is 05-SYSTEM-BIN-LOCATIO-S1-L1. Under 'Bin Location Properties', there are checkboxes for 'Inactive', 'Receiving Bin Location', and 'Exclude from Automatic Issue'. The 'Item Weight' field is highlighted with a yellow box and labeled 'Information only'. The 'Maximum Weight' field is highlighted with a yellow box and labeled 'Defined by the user', with the value '600kg' entered. Other fields include 'Description', 'No. of Items', 'Alternative Sort Code', 'Minimum Qty', 'Maximum Qty', 'Item Qty', 'No. of Batches/S', 'Bar Code', and 'Item Restrictions'.

Look at the fields *Item Weight* and *Maximum Weight* in the *Bin Location Master Data*. The *Item Weight* field is an information field showing the total weight of the items currently stored in the bin location. This total weight is a calculation of the quantity of each item in the bin location multiplied by the weight defined for the inventory UoM of the item. The *Maximum Weight* field is the maximum weight allowed for this bin location. This information is entered by the user.

Note:

- If no value is entered in the *Maximum Weight* field, then there will be no weight restriction applied to this bin location.
- In addition, if a weight restricted bin location contains an item without weight definition then this item will not be taken into account in the weight calculation of the bin location.

Exceeding Maximum Weight

- Warnings are provided for manual allocation only
- When using automatic allocation, the allocation is done even for a fraction of an item

[illegible]

- So what happens when an allocation is about to occur? For example when entering a *Goods Receipt PO* document and the maximum weight of a bin location is about to be exceeded.
- When automatic allocation is done – no alert is given. The system allocates up to the maximum weight allowed. This may result in partial allocation and even an allocation of a fraction of a unit.
- Alert is given for manual allocation only.
- In manual allocation, when entering the *Bin Location Allocation – Issue* window, and the allocation is about to exceed from the defined weight, an alert message is prompted as shown in the image.
- The user can then choose to allocate up to the maximum weight or to ignore the message.

Summary



Here are some key points:

- Weight can be managed for the item inventory UoM.
- The *Item Weight* field in the *Bin Location Master Data*, indicates the total weight of items currently stored in the bin location.
- The Maximum weight field indicates the weight limitation of the bin location.
- A weight restriction can be set for automatic and manual allocation, based on the bin location current weight and weight limitation.

Here are some key points to take away from this course:

- Weight can be managed for the item inventory UoM.
- The Item Weight field in the Bin Location Master Data, indicates the total weight of items currently stored in the bin location.
- The Maximum weight field indicates the weight limitation of the bin location.
- A weight restriction can be set for automatic and manual allocation, based on the bin location current weight and weight limitation.

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